Roll No. .....

TCE–403

B. TECH. (CIVIL ENGINEERING)

(FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

ADVANCED SURVEYING

Time : 1½ Hours

Maximum Marks : 50

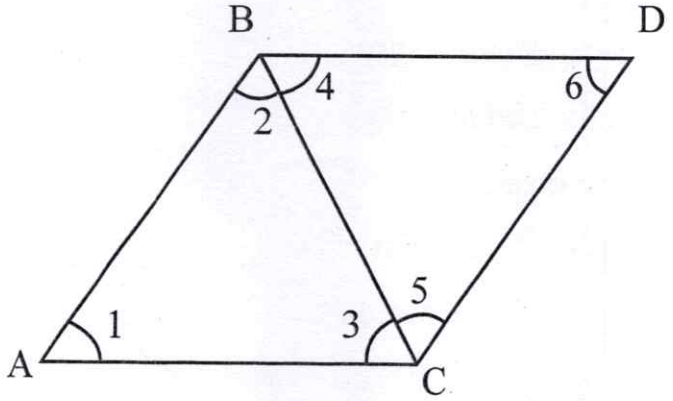
Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss the necessity of Control Surveying and differentiate between Geodetic and Plane Surveying. (CO1)

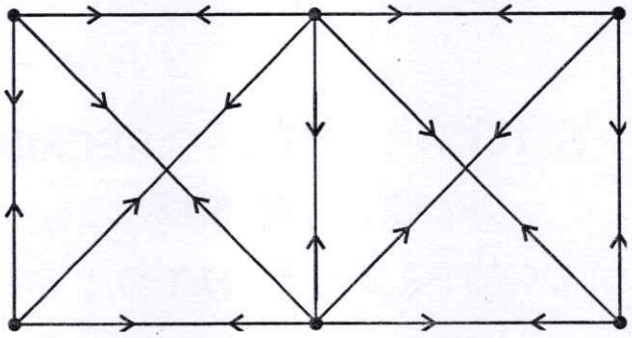
OR

(b) What is triangulation ? Also calculate the azimuths ($\theta_{AB}, \theta_{AC}, \theta_{BC}, \theta_{BD}$, and θ_{CD}) for all the lines in the triangulation system : (CO1)



P. T. O.

2. (a) List the draw basic triangulation figures. Explain different arrangements of triangulation figures with diagrams. Calculate n, n', s, s', and (D – C)/D : (CO1)



OR

(b) List all the instruments required to carry out a triangulation survey. Elaborate on the criteria for selection of triangulation stations. (CO1)

3. (a) Two triangulation stations A and B, 70 km apart have elevations of 370 m and 450 m respectively. The intervening ground has a uniform elevation of 320 m. Find the height of signal required at B if the line of sight has to pass at least 3 m above the ground at all points. (CO1)

OR

(b) Explain towers and satellite station along their purpose in a triangulation survey. Draw a well labelled diagram of a tower. (CO1)

4. (a) Define the following terms : (CO2)

(i) Independent quantity

(ii) Conditioned quantity

(iii) Direct observation

(iv) Indirect observation

(v) Weight of observation

OR

(b) In carrying a line of levels across a river, the following eight readings were taken with a level under identifiable conditions : (CO2)

2.322, 2.346, 2.352, 2.306, 2.312, 2.300, 2.306, 2.326

Calculate :

(i) the probable error

(ii) the probable error of the mean

5. (a) What are the three kinds of errors in measurement ? Explain consequences for each of them it undetected. (CO2)

OR

(b) An angle was measured by different persons and the following are the values : (CO2)

Angle	Number of measurements
65°30' 10"	2
65°29' 50"	3
65°30'00"	3
65°30'20"	4
65°30'10"	3

Find the most probable value of the angle.

60

Roll No. .....

TCE–402

B. TECH. (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

STRUCTURAL ANALYSIS—I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Determine all the member forces in the truss shown in figure 1 by the method of joints. (CO1)

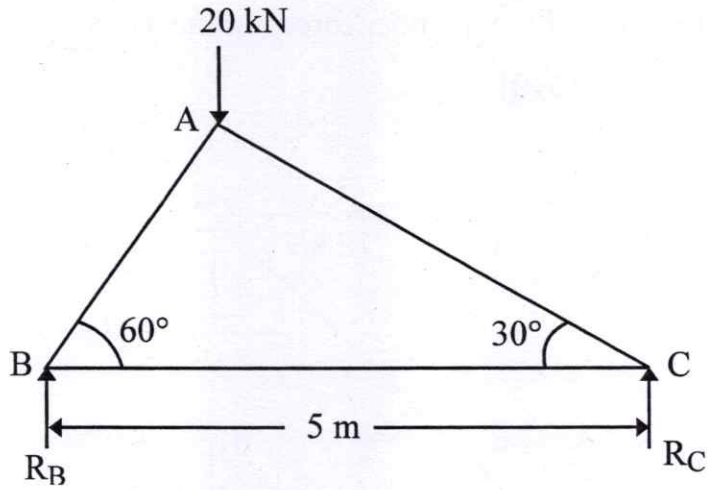


Fig. 1

OR

(b) Determine all the member forces in the truss shown in figure 1 by the method of Tension Coefficients. (CO1)

P. T. O.

2. (a) Determine all the member forces in the truss shown in figure 2 by the method of joints : (CO1)

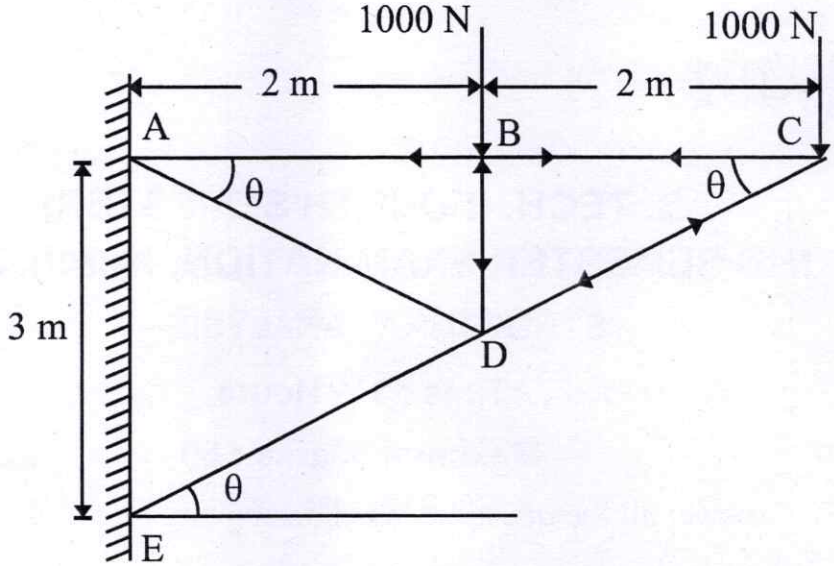


Fig. 2

OR

(b) Determine all the member forces in the shown in figure 3 by the method of Tension Coefficients. (CO1)

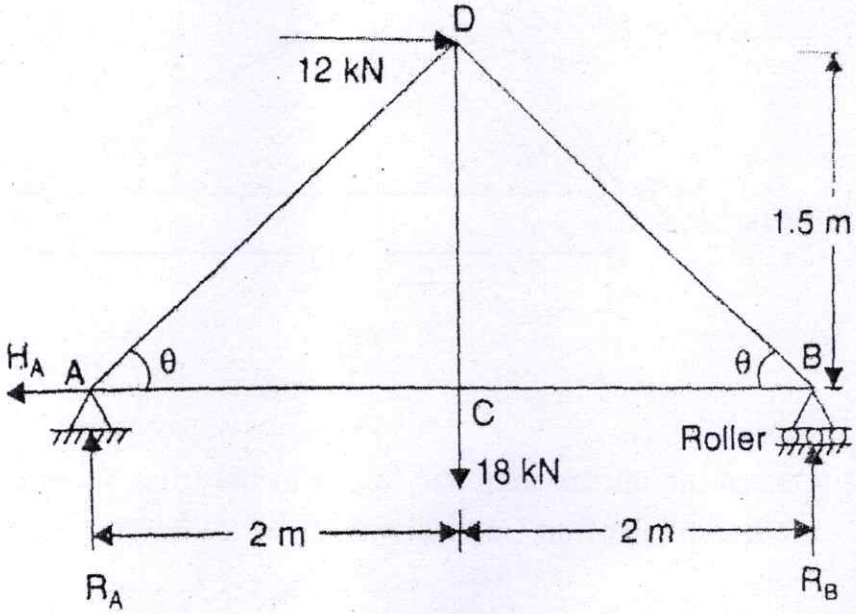


Fig. 3

3. (a) Using moment area method, determine the mid-span deflection in a simply supported beam of span 'L', when it is subjected to a udl throughout the span. (CO1)

OR

(b) A cantilever of length L is carrying a point load of W kN at a distance of 'a' from the fixed end. Calculate slope at the free end and deflection at the free end. (CO2)

4. (a) A simply supported beam AB of span 4 m carries a point load of 100 kN at its center C. The values of 1 for the left half is 1×10^8 mm^4 and for the right half portion 1 is 2×10^8 mm^4. Find the slopes at the two supports and deflection under the load using conjugate beam method. Take $E = 2 \times 10^5$ N/mm^2. (CO2)

OR

(b) A cantilever of length 3 m carries a point load of 10 kN at a distance of 2 m from the free end. If $E = 2 \times 10^5$ N/mm^2 and $I = 1 \times 10^8$ mm^4, find the slope and deflection at the free end using conjugate beam method. (CO2)

5. (a) Compare the strain energies of the following two shafts subject to the same maximum shear stress in torsion : (CO2)

(i) A hollow shaft having outer diameter n times the inner diameter.

(ii) A solid shaft

Masses, lengths and materials of the two shafts are the same.

OR

(b) A beam of length 1 simply supported at the ends is loaded with a point load W at a distance 'a' from one end. Assuming that the beam has constant cross-section with moment of inertia I and Young's modulus E, find the strain energy of beam and hence find the defection under the load. Stain energy due to shearing is neglected. (CO2)

60

Roll No. .....

TCE–405

B. TECH. (CIVIL ENGINEERING)

(FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

GEOTECHNICAL ENGINEERING—I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the following soil type : (CO1)

(i) LOESS

(ii) BENTONITE

(iii) PEAT

(iv) VARVED CLAY

OR

(b) For a given soil, the following are known : (CO1)

(i) Percentage passing No. 4 sieve (4.75 mm) = 70

(ii) Percentage passing No. 200 sieve (0.075 mm) = 30

(iii) Liquid limit = 33

(iv) Plastic limit = 12

P. T. O.

Classify the soil using the Unified soil classification system. Give the group symbol and the group name.

2. (a) Develop the following relationships : (CO1)

(i) $wG_s = Se$

(ii) Saturated unit weight in terms of water content, void ratio

(iii) Between void ratio and Porosity

(iv) Dry unit weight, bulk unit weight and water content

OR

(b) A proposed embankment fill requires 8000 m^3 of compacted soil. The void ratio of the compacted fill is specified as 0.7. Four borrow pits are available as described in the following table, which lists the respective void ratios of the soil and the cost per cubic meter for moving the soil to the proposed constructions site. Make the necessary calculations to select the pit from which the soil should be bought to minimize the cost. Assume Gs to be the same at all pits. (CO1)

Borrow pit	Void ratio	Cost ($/m^3)
A	0.82	8
B	1.1	5
C	0.90	9
D	0.78	12

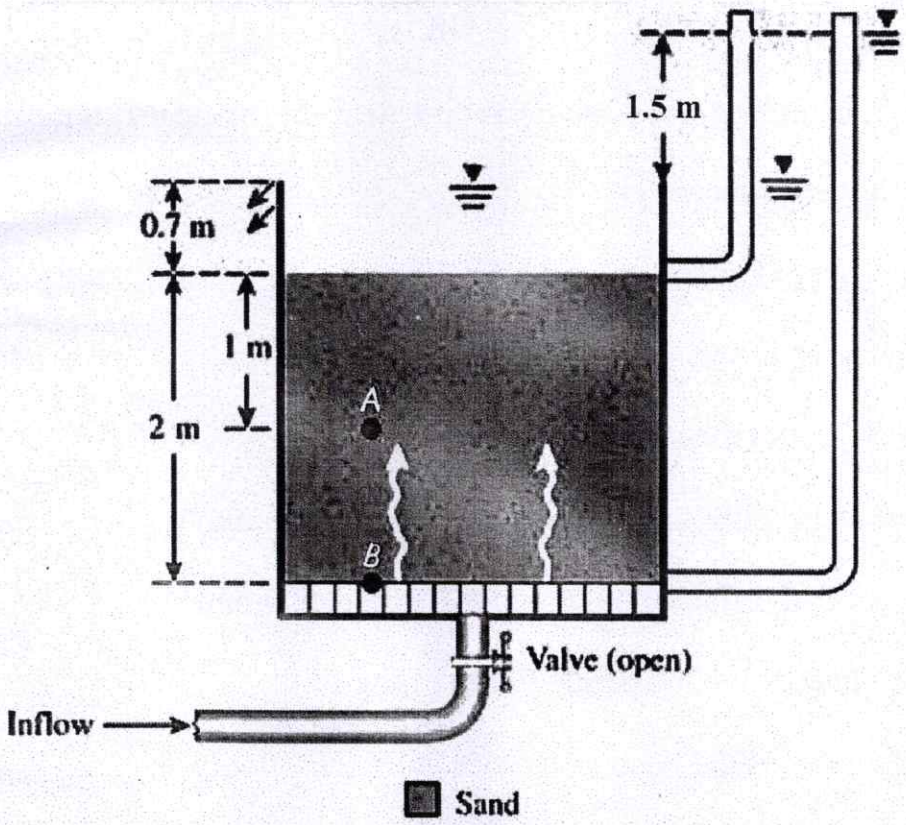
3. (a) Develop expression for estimating the coefficient of permeability of layered soil : (CO2)

(i) When flow is parallel to stratification

(ii) When flow is normal to stratification

OR

(b)



Consider the upward flow of water through a layer of sand in a tank as shown in Figure. For the sand, the following are given : (CO2)

void ratio (e) = 0.52 and specific gravity of solids = 2.67.

(i) Calculate the total stress, pore water pressure, and effective stress at points A and B.

(ii) What is the upward seepage force per unit volume of soil ?

4. (a) Explain the step by step procedure of determining the specific Gravity of soil solids using pycnometer method. Develop expression for obtaining specific gravity of solids in terms of measurement made in pycnometer test. (CO1)

OR

(b) For a falling-head permeability test at standard temperature, the following data are given : (CO1)

(i) Length of soil specimen = 700 mm

(ii) Area of the soil specimen = 20 cm^2

(iii) Area of the standpipe = 1.05 cm^2

(iv) Head difference at time t = 0 min is 800 mm

(v) Head difference at time t = 8 min is 500 mm

(I) Determine the permeability of the soil.

(II) What is the head difference at time t = 6 min ?

5. (a) Discuss the following : (CO2)

(i) Capillarity in Soil

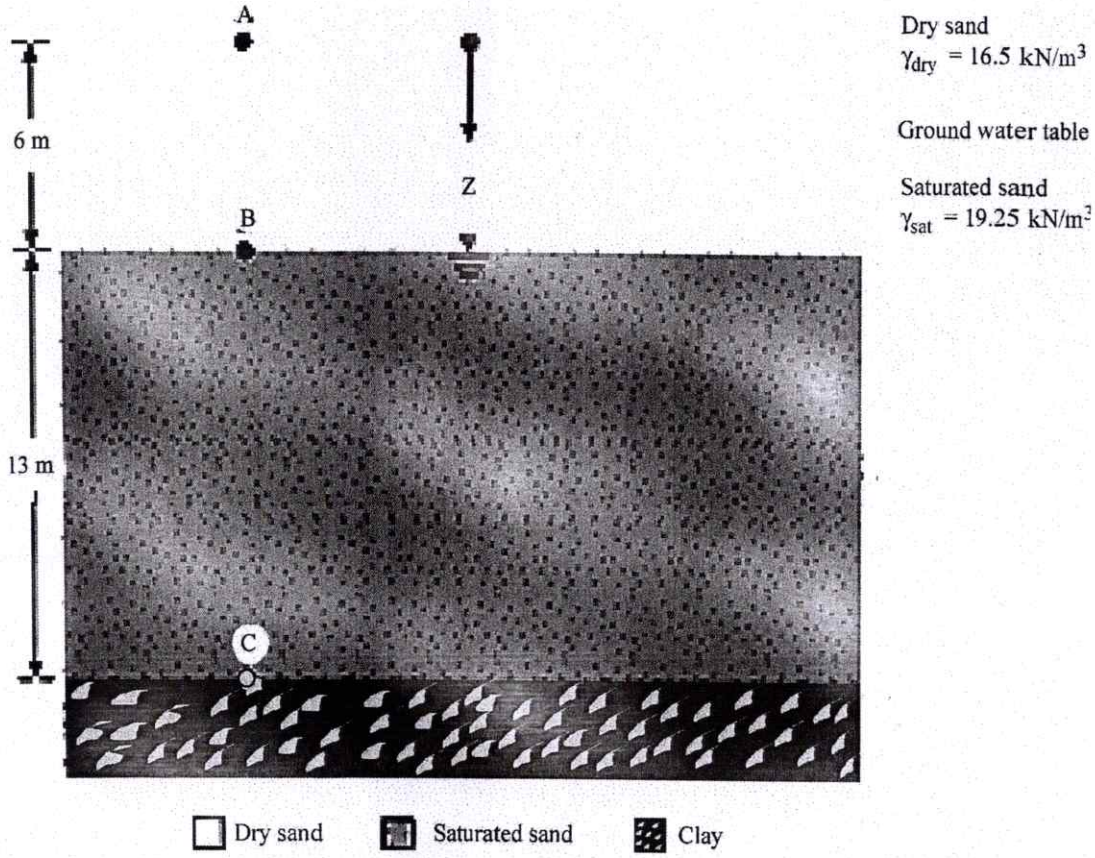
(ii) Homogeneous, isotropic and non-homogeneous soils

(iii) Critical hydraulic gradient at exit

(iv) Granular filter design philosophy

OR

(b)



For the given soil profile, estimate the amount of total stress, pore pressure and effective stress at points A, B and C. (CO2)

Roll No. .....

TCE-401

B. TECH. (CIVIL ENGINEERING)

(FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

HYDRAULICS AND HYDRAULIC MACHINES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) State the conditions under which the rectangular section of a open channel will be most efficient. Derive these conditions. (CO1)

OR

(b) An open channel of most economical section of trapezoidal shape is required to give a maximum discharge of 20.20 m^3/s of water. The slope of the channel bottom is 1 in 2500. Taking Chezy's constant C = 60 in Chezy's equation, determine the dimensions of the cross-section. (CO1)

2. (a) What is specific energy ? Discuss its relation with depth and differentiate different types of flow with the help of specific energy diagram. (CO1)

P. T. O.

OR

(b) A trapezoidal channel excavated in earth has to carry a discharge of 5 m^3/s at a velocity of 0.75 m/s. The channel is 5 m wide at the base and side slope 1 : 1. At what slope should it be laid ? Take Manning's $n = 0.02$. (CO1)

3. (a) Obtain the differential equation for water surface slope. List all the assumptions. (CO2)

OR

(b) A rectangular channel 4 m wide conveys a discharge of 14 m^3/s at a depth of 1.75 m. If the bed is raised by 0.20 m, will the water surface in the channel upstream be disturbed ? Find the depth of flow over the raised bed and fall in the water surface. (CO2)

4. (a) State the classification of channels on the basis of bed slope. Draw the water surface profile for steep bed slope. (CO2)

OR

(b) A tidal river flowing towards the sea at a velocity of 1.50 m/s with a depth of flow of 2 m, encounters an incoming tide which results in an increased flow depth of 4.80 m. The tidal bore is moving upstream. Find the velocity of the tidal bore and the velocity of water after the passage of the bore. (CO2)

5. (a) With the help of a neat diagram, classify hydraulic jump based on Froude's number. (CO2)

OR

(b) In a rectangular channel of 0.5 m width, a hydraulic jump occurs at a point where depth of flow is 0.15 m and Froude's no. is 2.5. Determine the specific energy, critical depth, sequent depth and head loss. (CO2)

Roll No. .....

TCE–406

B. TECH. (CE) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

ENGINEERING GEOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Write a short note on Moh's scale of hardness with examples. (CO1)

OR

(b) Enlist and explain the subdivisions of Geology. (CO1)

2. (a) Draw and explain any *five* different forms or structure of minerals. (CO1)

OR

(b) Describe the internal structure of the Earth with the help of a neat diagram. (CO1)

3. (a) Compare and contrast the following pairs : (CO1, CO2)
   (i) Lava and Magma
   (ii) Sills and Dykes
   (iii) Concordant and Discordant bodies

OR

(b) What are the various requirements of a good building stone ? (CO1, CO2)

P. T. O.

4. (a) Distinguish between igneous, sedimentary and metamorphic rocks. (CO2)

OR

(b) What is metamorphism ? Discuss the various agents of metamorphism along with some examples. (CO2)

5. (a) Define the following terms : (CO2)

(i) Outcrop

(ii) Bedding or Stratification

(iii) Dip

OR

(b) Briefly explain the classification of Igneous rocks. (CO2)